

# YUN-TING CHENG

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## PROFESSIONAL EXPERIENCE

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### California Institute of Technology/Jet Propulsion Laboratory

Research Scientist

Jan 2024 – present

### Jet Propulsion Laboratory/California Institute of Technology

Postdoctoral Researcher

Oct 2021 – Jan 2024

## EDUCATION

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### California Institute of Technology

Ph.D. in Physics (Advisor: Prof. Jamie Bock)

Jun 2021

Thesis: [Cosmology and Astrophysics with Intensity Mapping](#)

M.S. in Physics

Jun 2019

### National Taiwan University

B.S. in Physics

Jun 2014

## RESEARCH INTERESTS

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- Development of analysis algorithms in galaxy surveys
- Characterization of observational systematics of large-scale structure surveys
- Photometric redshift inference and validation
- Optimal extraction of spectral-line intensity mapping signals
- Study of the extragalactic background

## RESEARCH EXPERIENCE

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### California Institute of Technology / Jet Propulsion Laboratory

Pasadena, CA

#### *Research Scientist*

Jan 2024 – present

- SPHEREx Mission
  - Characterizing the photometric redshift estimation and systematics
  - Optimizing the multi-tracer analysis for SPHEREx cosmology science
  - Modeling the extragalactic background and line intensity mapping signal for SPHEREx
- Photometric Redshift with Clustering Statistics (collaborators: E. Huff, K. Markovic, I. Szapudi)
  - Developing a novel photometric redshift technique with clustering information

#### *Postdoctoral Researcher* (Advisor: Dr. Olivier Doré)

Oct 2021 – Jan 2024

- 3D Light Cones Cosmology (collaborators: B. Wandelt, O. Doré, T.-C. Chang)
  - Developing a Bayesian data-driven technique to optimally constrain cosmology with spectral imaging data
- Galactic Extinction Modeling (collaborators: B. Hensley, O. Doré, T.-C. Chang)
  - Building Galactic dust extinction model from multi-tracer datasets
- Extragalactic Radio Dipole (collaborators: A. Lidz, T.-C. Chang)
  - Modeling the cosmic dipole signal from extragalactic radio sources
- Intra-halo Light Signal in the Extragalactic Background (collaborator: J. J. Bock)
  - Modeling non-linear clustering and intra-halo light in the near-infrared background

#### *Graduate Research Assistant* (Advisor: Prof. Jamie Bock)

Sep 2015 – Jun 2021

- Line Intensity Mapping
  - Developing analysis algorithms to tackle line confusion in line intensity mapping surveys
  - Establishing formalism of optimal mapping strategies for large-scale structure surveys
  - Modeling galaxy-intensity mapping cross-correlation for SPHEREx
- CIBER (Cosmic Infrared Background Experiment)
  - Building CIBER analysis pipeline and characterizing observational systematic effects
  - Studying intra-halo light with stacking analysis on CIBER images
- TIME (Tomographic Ionized Carbon Intensity Mapping Experiment)
  - Simulating signal and foregrounds for TIME analysis pipeline
  - Developing foreground mitigation techniques
  - Analyzing TIME instrument data
  - Participating in instrument deployment at the ARO 12m telescope

**Academia Sinica of Astronomy and Astrophysics (ASIAA)****Research Assistant** (Advisor: Tzu-Ching Chang)

- Developing foreground mitigation techniques for line intensity mapping

**Summer Student** (Advisors: Sheng-Yuan Liu, Yu-Nung Su, I-Ta Hsieh)

- Modeling the starless core with radiative transfer

Taipei, Taiwan

May 2014 – Jul 2015

Jul 2013 – Aug 2013

**GRANTS**

- **NASA Astrophysics Data Analysis Program (ADAP)** 2025 – 2027  
Title: Detecting Baryonic Acoustic Oscillations in the WISE galaxy sample  
Role: Co-I (PI: Katarina Markovic, JPL)
- **NASA Astrophysics Data Analysis Program (ADAP)** 2025 – 2027  
Title: Cosmology with Juno: Extracting the Cosmic Microwave Background Dipole at Low Frequencies and the Nature of the Diffuse Radio Background  
Role: Co-I (PI: Tzu-Ching Chang, JPL)

**TECHNICAL SKILLS**

- **Statistical Tools:** Bayesian statistics, Markov Chain Monte Carlo, Fisher analysis, sparse reconstruction, convex optimization, Machine Learning
- **Programming Languages:** Python (JAX, Astropy, emcee, Pandas, scikit-learn, TensorFlow, Keras, seaborn), SQL, IDL, Matlab, C++, Fortran, LaTeX
- **Instrumentation:** SOLIDWORKS, trained in machine shop techniques

**STUDENT MENTORING**

- Jean Choppin de Janvry, Paris-Saclay University, Caltech Visiting Master Student Sep 2024 – Mar 2025  
Topic: Photometric Redshift for SPHEREx
- Gemma (Zhaoyu) Huai, Caltech Physics Graduate Student Jun 2024 – present  
Topic: SPHEREx Cosmology
- Jui-Kuan Chan, National Taiwan University, Caltech SURF Program Jun 2024 – Aug 2024  
Topic: SPHEREx Photometric Redshift With SOM
- Kailai Wang, Cornell University, JPL SURF Program Jun 2023 – Aug 2023  
Topic: Bayesian Multi-line Intensity Mapping
- Abby Williams, NYU/Caltech, Caltech Post-baccalaureate Student Jun 2023 – Jan 2026  
Topic: Small-scale Nonlinear Effects in Cross Correlations

**SERVICE AND OUTREACH**

- The Chinese University of Hong Kong Study Tour Program, Lecturer Jul 2025
- Caltech Cosmology Journal Club, Co-organizer Sep 2022 – present
- 237th AAS Meeting, Oral Session Chair Jan 2021
- Gabrielino High School, CA, Physics In-Class Activities Leader Jan 2020 – Sep 2022
- 233rd AAS Meeting, Poster Judge Jan 2019
- Journal referee: *Astrophysical Journal (ApJ)*, *Astrophysical Journal Letters (ApJL)*, *Journal of Cosmology and Astroparticle Physics (JCAP)*, *Monthly Notices of the Royal Astronomical Society (MNRAS)*, *Astronomy & Astrophysics (A&A)*

**AWARDS AND HONORS**

- Balzan Cosmological Studies Travel Award Oct 2022
- Taiwan-Caltech Ministry of Education Fellowship Sep 2015 – Aug 2019
- Dean's Award of College of Science, National Taiwan University Jun 2014

## REFERENCES

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- **James J. (Jamie) Bock**  
Professor, California Institute of Technology/Jet Propulsion Laboratory  
[jjb@astro.caltech.edu](mailto:jjb@astro.caltech.edu)
- **Tzu-Ching Chang**  
Research Scientist, Jet Propulsion Laboratory/California Institute of Technology  
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- **Olivier P. Doré**  
Research Scientist, Jet Propulsion Laboratory/California Institute of Technology  
[olivier.p.dore@jpl.nasa.gov](mailto:olivier.p.dore@jpl.nasa.gov)/[odore@caltech.edu](mailto:odore@caltech.edu)
- **Abigail T. Crites**  
Assistant Professor, Cornell University  
[atc72@cornell.edu](mailto:atc72@cornell.edu)
- **Benjamin D. Wandelt**  
Professor, Johns Hopkins University  
[wandelt@jhu.edu](mailto:wandelt@jhu.edu)

## PRESENTATIONS

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### Invited Conference Talks and Lectures

- KITP Program: CMB x LSS ([video](#)) Santa Barbara, CA, Feb, 2026
- Cosmoglobe Bayesian Cosmological Data Analysis Course Oslo, Norway, Sep, 2025
- Cosmology on the Steep Rise Workshop Sexten, Italy, Feb, 2025
- A new era with Line Intensity Mapping Workshop Heidelberg, Germany, Dec, 2024
- Line Intensity Mapping Workshop MPA, Garching, Germany, Apr, 2023

### Invited Institution Colloquia and Seminars

- Caltech IPAC Seminar Pasadena, CA, Oct, 2025
- ASIAA Colloquium Taipei, Taiwan, Dec, 2024
- University of Bonn/MPIfRA Colloquium Bonn, Germany, Dec 2024
- KIPAC Seminar Stanford, CA, Nov, 2024
- ASIAA Seminar Taipei, Taiwan, Oct, 2023
- LAM Cafe Club LAM, Marseille, France, Dec, 2022
- IAP Universe Seminar IAP, Paris, France, Nov, 2022
- NYU CCPP Seminar NYU, NY, Aug, 2022
- ICAP seminar (virtual) IAP, Paris, France, Jun, 2022
- ASIAA Seminar ([video](#)) Taipei, Taiwan, May, 2022
- IRSIG Webminar (virtual), Oct, 2021
- ASIAA Seminar (virtual) Taipei, Taiwan, Mar, 2021
- UChicago KICP Seminar (virtual) Chicago, IL, Jan, 2021
- Berkeley BCCP Seminar (virtual) Berkeley, CA, Dec, 2020
- CCA Flatiron Institute Cosmology Group Meeting (virtual) CCA, NY, Oct, 2020
- OSU CCAPP Seminar (virtual) OSU, OH, Oct, 2020
- JHU Cosmology/GW Journal Club (virtual) JHU, MD, Oct, 2020
- UPenn Astronomy Seminar (virtual) UPenn, PA, Sep, 2020
- CCA Flatiron Institute Lunch Talk (virtual) CCA, NY, Sep, 2020
- ASIAA Seminar Taipei, Taiwan, Sep, 2018

### Contributed Talks

- Infrared Spectroscopy from Space Pasadena, CA, Oct, 2025
- Line Intensity Mapping 2025 Annecy, France, Jun, 2025
- PRIMA and the Future of Far-Infrared Science Pasadena, CA May, 2025
- Roman Community Workshop ([video](#)) Pasadena, CA Jul, 2024
- Line Intensity Mapping 2024 Workshop ([video](#)) UIUC, IL, Jun, 2024
- Diffuse Cosmic Backgrounds and the Low Surface Brightness Universe Aspen, CO, Apr, 2024
- Caltech ObsCos Seminar Pasadena, CA Feb, 2024
- 243rd AAS Meeting New Orleans, LA, Jan, 2024
- Pan-Ex Meeting (virtual), Nov, 2023
- JPL Postdoc Seminar JPL, Pasadena, CA, Nov, 2023
- Probing the Universe at High Resolution Conference Taipei, Taiwan, Nov, 2023
- LAM CONCERTO Working Group Meeting LAM, Marseille, France, Dec, 2022
- Caltech ObsCos Seminar Pasadena, CA Oct, 2022
- Columbia Cosmology Group Seminar Columbia, NY, Aug, 2022
- Cosmology from Home ([video](#)) (virtual), Jun, 2022
- Cross Correlations with CHORD Workshop (virtual), Oct, 2021
- SUBLIME Workshop (virtual), Oct, 2021
- KICP Line Intensity Mapping Workshop (virtual) Chicago, IL, Jul, 2021
- Caltech ObsCos Seminar (virtual) Pasadena, CA Feb, 2021
- 237th AAS Meeting (virtual), Jan, 2021
- Caltech ObsCos Seminar (virtual) Pasadena, CA Sep, 2020

- CCAT-prime Science Working Group Meeting (virtual) Cornell, NY, Sep, 2020
- Caltech ObsCos Seminar Pasadena, CA Feb, 2020
- L2S2 : Lines in the Large Scale Structure Conference Marseille, France, Jul, 2019
- Caltech ObsCos Seminar Pasadena, CA Jun, 2019
- Caltech ObsCos Seminar Pasadena, CA May, 2019
- 233rd AAS Meeting Seattle, WA, Jan, 2019
- Taiwanese Theoretical Astrophysics Workshop Taipei, Taiwan, Sep, 2018
- Caltech ObsCos Seminar Pasadena, CA Jun, 2018
- Cosmological Signals from Cosmic Dawn to the Present Aspen, CO, Feb, 2018
- Caltech ObsCos Seminar Pasadena, CA Dec, 2017
- Caltech ObsCos Seminar Pasadena, CA Nov, 2016
- Caltech ObsCos Seminar Pasadena, CA Jun, 2016
- Opportunities and Challenges in Intensity Mapping Workshop KIPAC, CA, Mar, 2016
- ASROC Annual Meeting (Taiwanese Astronomical Society) Ilan, Taiwan, May, 2015

## PUBLICATIONS

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See [ADS](#), [Google Scholar](#), and [INSPIRE](#) for the complete publication list

### First-author (11 published / accepted)

1. *Feature Intensity Mapping: Polycyclic Aromatic Hydrocarbon Emission from All Galaxies Across Cosmic Time*  
**Y.-T. Cheng**, B. S. Hensley, T. S.-Y. Lai, 2025, ApJ, 993, 30, [ADS](#), [[arXiv:2506.13863](#)]
2. *Mapping Galactic Dust Emission and Extinction with HI, HII, and H<sub>2</sub>*  
**Y.-T. Cheng**, B. S. Hensley, T.-C. Chang, & O. Doré, 2025, ApJ, 985, 15, [ADS](#), [[arXiv:2411.12801](#)]
3. *Bayesian Multi-line Intensity Mapping*  
**Y.-T. Cheng**, K. Wang, B. D. Wandelt, T.-C. Chang, & O. Doré, 2024, ApJ, 971, 159, [ADS](#), [[arXiv:2403.19740](#)]
4. *Is the Radio Source Dipole from NVSS Consistent with the CMB and  $\Lambda$ CDM?*  
**Y.-T. Cheng**, T.-C. Chang, & A. Lidz, 2023, ApJ, 965, 32, [ADS](#), [[arXiv:2309.02490](#)]
5. *Data-driven Cosmology from Three-dimensional Light Cones*  
**Y.-T. Cheng**, B. D. Wandelt, T.-C. Chang, & O. Doré, 2023, ApJ, 944, 151, [ADS](#), [[arXiv:2210.10052](#)]
6. *Near-infrared Extragalactic Background Light Fluctuations on Nonlinear Scales*  
**Y.-T. Cheng** & J. J. Bock, 2022, ApJ, 940, 115, [ADS](#), [[arXiv:2207.13712](#)]
7. *Cosmic Near-Infrared Background Tomography with SPHEREx Using Galaxy Cross-Correlations*  
**Y.-T. Cheng** & T.-C. Chang, 2022, ApJ, 925, 136, [ADS](#), [[arXiv:2109.10914](#)]
8. *Probing Intra-Halo Light with Galaxy Stacking in CIBER Images*  
**Y.-T. Cheng** et al. (CIBER Collaboration), 2021, ApJ, 919, 69, [ADS](#), [[arXiv:2103.03882](#)]
9. *Phase-Space Spectral Line De-confusion in Intensity Mapping*  
**Y.-T. Cheng**, T.-C. Chang, & J. J. Bock, 2020, ApJ, 901, 142, [ADS](#), [[arXiv:2005.05341](#)]
10. *Optimally Mapping Large-Scale Structures with Luminous Sources*  
**Y.-T. Cheng**, R. de Putter, T.-C. Chang, & O. Doré, 2019, ApJ, 877, 86, [ADS](#), [[arXiv:1809.06384](#)]
11. *Spectral Line De-Confusion in an Intensity Mapping Survey*  
**Y.-T. Cheng**, T.-C. Chang, J. J. Bock, C. M. Bradford, & A. R. Cooray, 2016, ApJ, 832, 165, [ADS](#), [[arXiv:1604.07833](#)]

### Contributing author

1. *Euclid preparation. Decomposing components of the extragalactic background light using multi-band intensity mapping cross-correlations*  
Y. Cao, et al., 2026, submitted, [ADS](#), [[arXiv:2601.21111](#)]
2. *SPHEREx Pre-Perihelion Mapping of H<sub>2</sub>O, CO<sub>2</sub>, and CO in Interstellar Object 3I/ATLAS*  
C. M. Lisse, et al., 2025, ApJL submitted, [ADS](#), [[arXiv:2512.07318](#)]
3. *TIME Commissioning Observations: I. Mapping Dust and Molecular Gas in the Sgr A Molecular Cloud Complex at the Galactic Center*  
S. F. Yang, et al., 2025, ApJ submitted, [ADS](#), [[arXiv:2511.09473](#)]
4. *The Redshifts from 122 Bands: Comparative Redshift Forecast for Low-Resolution Spectra from SPHEREx and 7-Dimensional Sky Survey (7DS)*  
J. Bae, et al., 2025, A&A, 706, A347 [ADS](#), [[arXiv:2512.24537](#)]
5. *The SPHEREx Satellite Mission*  
Bock, et al., 2025, ApJ accepted, [ADS](#), [[arXiv:2511.02985](#)]
6. *Simulating Spectral Confusion in SPHEREx Photometry and Redshifts*  
Z. Huai, et al., 2025, ApJ accpeted, [ADS](#), [[arXiv:2510.01410](#)]
7. *The SPHEREx Sky Simulator: Science Data Modeling for the First All-Sky Near-Infrared Spectral Survey*  
B. P. Crill, et al., 2025, ApJS, 281, 10, [ADS](#), [[arXiv:2505.24856](#)]
8. *The Potential of the SPHEREx Mission for Characterizing PAH 3.3  $\mu$ m Emission in Nearby Galaxies*  
E. Zhang, A. L. Faisst, et al., 2025, ApJ 992, 32, [ADS](#), [[arXiv:2503.21876](#)]

9. *CIBER 4th flight fluctuation analysis: Measurements of near-IR auto- and cross-power spectra on arcminute to sub-degree scales*  
R. M. Feder, et al., 2025, ApJ, 994, 30, [ADS](#), [[arXiv:2501.17933](#)]
10. *CIBER 4th flight fluctuation analysis: Pseudo-power spectrum formalism, improved source masking and validation on mocks*  
R. M. Feder, et al., 2025, ApJ, 991, 87, [ADS](#), [[arXiv:2501.17932](#)]
11. *Inferred Measurements of the Zodiacal Light Absolute Intensity through Fraunhofer Absorption Line Spectroscopy with CIBER*  
P. M. Korngut, et al., 2022, ApJ, 926, 133, [ADS](#), [[arXiv:2104.07104](#)]
12. *Probing Cosmic Reionization and Molecular Gas Growth with TIME*  
G. Sun, T.-C. Chang, et al., 2021, ApJ, 915, 33, [ADS](#), [[arXiv:2012.09160](#)]
13. *Superresolution Reconstruction of Severely Undersampled Point-spread Functions Using Point-source Stacking and Deconvolution*  
T. Symons, M. Zemcov, et al., 2021, ApJS, 252, 24, [ADS](#), [[arXiv:2102.01094](#)]
14. *Hafnium Films and Magnetic Shielding for TIME, A mm-Wavelength Spectrometer Array*  
J. Hunacek, et al., 2018, JLTP, 193, 893, [ADS](#)
15. *A Foreground Masking Strategy for [C II] Intensity Mapping Experiments Using Galaxies Selected by Stellar Mass and Redshift*  
G. Sun, L. Monceli, M. P. Viero, et al., 2018, ApJ, 856, 107, [ADS](#), [[arXiv:1601.10095](#)]
16. *Design and fabrication of TES detector modules for the TIME-Pilot [C II] Intensity Mapping Experiment*  
J. Hunacek, et al., 2016, JLTP, 184, 733, [ADS](#)

## White paper / review paper contributions

1. *PRIMA General Observer Science Book Volume 2*  
A. Moullet, et al., 2025, [ADS](#), [[arXiv:2511.10927](#)]  
Case #17 – Probing PAH from All Galaxies Across Redshifts – PAH Intensity Mapping with PRIMA [[ADS](#)]
2. *PRIMA General Observer Science Book*  
A. Moullet, et al., 2023, [ADS](#), [[arXiv:2310.20572](#)]  
Case #9: Probing the history of cosmic star formation, black hole growth, and metallicity/dust evolution with line-intensity mapping
3. *Tomography of the Cosmic Dawn and Reionization Eras with Multiple Tracers*  
T.-C. Chang, et al., 2019, Astro2020 White Paper, [ADS](#), [[arXiv:1903.11744](#)]
4. *Line-Intensity Mapping: 2017 Status Report*  
E. D. Kovetz, M. P. Viero, et al., 2017, [ADS](#), [[arXiv:1709.09066](#)]

## Conference proceedings

1. *A status update on TIME: a mm-wavelength spectrometer designed to probe the Epoch of Reionization*  
A. Crites, et al., 2020, [SPIE](#), [114530G](#)
2. *Detector modules and spectrometers for the TIME-Pilot [CII] intensity mapping experiment*  
J. Hunacek, et al., 2016, [SPIE](#), [99140L](#)